

Forest Carbon Estimation: Confidence, Accuracy & Uncertainty

A methodology and limitations reference for the 180Climate Forest Carbon Pre-Feasibility Screening Tool

Prepared for	Document type	Version / date	Status
180Climate	Methodology & limitations reference	v1.0 · 23 June 2026	Indicative — screening-grade

Headline. 180Climate's carbon figures are deliberately reported as **ranges with a stated uncertainty band** — never a single number and never a single “% accuracy”. From free, open satellite data the tool produces an **indicative, screening-grade** estimate suitable for go/no-go decisions, not for financing or credit forecasting. Higher-resolution satellite data narrows the measurement error but does **not** resolve the largest source of uncertainty: the **baseline** (what would have happened anyway). The real step-change in confidence comes from **field measurement + an accredited methodology + third-party verification** — not from satellite resolution alone.

1. Purpose & scope

This report explains, honestly and in defensible terms, **how confident a user can be** in the carbon volumes the 180Climate tool estimates, what drives the uncertainty, and how that confidence changes when free satellite data is swapped for higher-resolution commercial data or upgraded with field measurement. It is written so that a technical reviewer — a rival developer, a project verifier, or a funder's advisor — can see that the tool's claims are grounded in established forest-carbon and remote-sensing practice.

The 180Climate tool is an **indicative pre-feasibility screening instrument**. Its job is to give a landowner or developer a fast, transparent first read on whether a forest concession is worth taking to a full feasibility study. It is explicitly **not** a registry-grade quantification tool, and nothing in it should be read as a guarantee of issuable carbon credits.

2. What the tool estimates

The headline figure is **avoided emissions** — the greenhouse gas, in tonnes of CO₂-equivalent (tCO₂e), that would be released if the forest were legally harvested under its permit, and that the project avoids by keeping the forest standing. In simplified form:

Avoided emissions = (emissions under the legal-harvest baseline) – (emissions with the project in place) – leakage – buffer/permanence deductions

The additionality basis is the **foregone legal harvest right** — the owner gives up a permitted, lawful right to clear or log. Consistent with the tool's methodology-agnostic design, the engine outputs a **transparent range** (a low and a high value) with the assumptions stated, and the AI narrative names the appropriate Verra family by permit type. It never converts these into a single, falsely precise figure.

3. The anatomy of uncertainty

A forest-carbon estimate is not one measurement; it is a **chain of estimates multiplied together**, and the errors compound. Critically, only some links in the chain are affected by satellite resolution. The single most important thing to understand is in the final column below.

Uncertainty component	What it is	Typical magnitude*	Improved by higher-res satellite?
Area / boundary	Size and edge of the forest parcel	Low (± 5 –15%)	Modestly — boundaries are usually known
Carbon density (biomass)	Tonnes of carbon per hectare	High (± 30 –50% fine-scale)	Yes — the main thing satellites improve
Baseline / counterfactual	What would have happened without the project	Very high — can dominate the result	No — this is methodological, not optical
Permanence	Risk the carbon is released later	Handled by 10–20% buffer pools	No
Leakage	Deforestation displaced elsewhere	Material, project-specific	No
Methodology / model choice	Allometric models, emission factors, flexibility in rules	Material; adds optionality bias	No

*Illustrative ranges synthesised from the remote-sensing literature; actual values vary by forest type, data, and method. Magnitudes are not additive — they propagate, and the dominant term governs the result.

The key insight. Satellite resolution mainly improves **carbon density** (and a little of **area**). It does **nothing** for the **baseline**, leakage, permanence, or methodology — and for issued credits the **baseline is usually the largest error of all** (Section 8). So “better satellites” cannot, on their own, take a screening estimate to high confidence on creditable volume.

4. Why we never quote a single “% confidence”

There is a real difference between two things people conflate:

- **A statistical confidence interval** — e.g. “X tCO₂e, with a 90% interval of [low, high]”. This is rigorous and is what serious practitioners report.
- **A single ‘accuracy %’** — e.g. “our estimate is 85% accurate”. This is not a well-defined quantity for a multi-component avoided-emissions estimate, and experienced reviewers treat it as a warning sign.

Accordingly, 180Climate reports a **range plus an uncertainty band** and a plain-language confidence tier — and deliberately avoids a single accuracy percentage. The industry's own framing for ‘how good is this number’ is the **IPCC Tier system** (Section 9) and standard error statistics such as %RMSE and R-squared — not a marketing-style accuracy figure.

5. Accuracy of free, open satellite data

The tool's free tier draws on the best openly available Earth-observation products. Each is strong for a specific purpose and weak for others:

Open data source	Type & resolution	Best used for	Key limitation
GFW / Hansen GFC	Optical, 30 m, annual since 2000	Tree-cover loss history, screening	Coarse; loss $\neq$ carbon; no degradation

Open data source	Type & resolution	Best used for	Key limitation
JRC Tropical Moist Forest	Optical, ~30 m, long time series	Disturbance & recovery, deforestation	Class-based; not biomass
RADD alerts	Radar (Sentinel-1), ~10 m, near-real-time	Recent loss, cloud-prone tropics	Detection, not carbon stock
ESA CCI Biomass	Modelled map, ~100 m	First-pass biomass density	Large pixel uncertainty (often >30–50%)
GEDI / ICESat-2	Spaceborne LiDAR, ~25 m footprints	Canopy height & biomass calibration	Sampled footprints, not wall-to-wall
Sentinel-2	Optical, 10 m, free	Boundary, land cover, change	Optical (cloud); indirect for biomass

On carbon density specifically, independent validation of global/free biomass products shows substantial error: global wall-to-wall maps report modest skill (for example, ~1 km products around R-squared 0.5–0.6 with root-mean-square error near 85–90 Mg/ha; some 30 m products lower), and fine-scale per-pixel uncertainty often runs **±30–50% or more**, exceeding 50% in low-biomass or heterogeneous areas (Santoro & Cartus / ESA CCI; Dubayah et al. / GEDI; multiple peer-reviewed validations).

What helps you: over a **large concession** (the tool's 20,000 ha+ minimum), per-pixel errors partly average out, so the **relative** uncertainty on the *total* biomass is smaller than the per-pixel figure. This is why free data is genuinely useful for **screening at scale** — but the baseline term (Section 8) does not average away.

6. Accuracy of higher-resolution & commercial data

Upgrading to airborne LiDAR, or commercial very-high-resolution optical/stereo (e.g. Planet, Maxar) and newer radar/LiDAR missions, materially tightens the **measurement** terms:

- **Biomass uncertainty** can fall to roughly **±15–20%** when calibrated with LiDAR. Peer-reviewed studies report %RMSE dropping from the low-30s to ~19–21% once LiDAR is added to radar/optical data.
- **Boundaries and small features** are resolved more precisely, helping with small or irregular plots.
- **Field-grade validation** still requires physically measuring and (for the strongest validation) destructively sampling trees — satellite estimates of any resolution are validated **indirectly**.

Honest ceiling. Higher resolution moves you from 'screening' to 'improved pre-feasibility' — but it is still a **desktop** estimate with a **modelled baseline** and **no field plots**. It does not, by itself, deliver high confidence on creditable volume. Claiming “now we are 90% accurate” after only a satellite swap would repeat the very mistake this report warns against.

7. The dominant uncertainty: the baseline

The evidence here is strong and recent, and it is the single most important point for credibility. Large-scale independent evaluations of issued forest-carbon credits have found systematic **over-crediting driven by the baseline** — not by the choice of satellite/forest-cover data:

Independent evaluation	Finding
West et al., 2023 (Science)	Emission reductions overstated for ~71% of evaluated REDD+ projects; net overestimation exceeding 17.6 million tCO ₂ e.

Independent evaluation	Finding
Guizar-Coutiño / Cambridge, 2023	Across 18 projects in five countries, crude historical/regional baselines inflated apparent conservation success.
Haya et al. (Berkeley) / CMW, 2023	The four methodologies behind almost all REDD+ credits showed large over-crediting; uncertainty lets participants pick credit-inflating assumptions.
Synthesis of 44 projects, 2026 (Nature Comms.)	Projects claimed in aggregate ~10.7× more avoided deforestation than independent estimates justify — driven by control-area selection and modelling, not by forest-cover data.
52-project study, 2025 (Science)	Only ~19% of projects met their reported emissions targets; on the order of one in eight tradable credits is supported by counterfactual analysis.

Why this matters for 180Climate. The 2026 synthesis is explicit that the ~10.7× discrepancy was **not** driven by the choice of forest-cover data. In other words, **no satellite — free or paid — fixes the baseline problem.** Confidence in *creditable* volume comes from a defensible, ideally ex-post and independently set baseline, an accredited methodology, and third-party verification. The 180Climate tool therefore presents avoided-emissions as an **indicative range** and routes the baseline question to a qualified methodology advisor — it does not pretend to settle it.

8. The confidence ladder (IPCC Tiers)

Reviewers think in IPCC Tiers. This is the credible way to describe where any given estimate sits, and it maps cleanly onto 180Climate's upgrade path:

Tier	Data basis	Typical biomass uncertainty	What it credibly supports
Tier 1 — Screening	Free satellite + default/global factors (the free tool)	±30–50% (tighter when aggregated)	Go/no-go screening; lead qualification
Toward Tier 2	Higher-res / LiDAR-calibrated, region-specific data	≈ ±15–20%	Improved pre-feasibility; investment scoping
Tier 3 — Registry-grade	Field plots + accredited methodology + independent verification (VVB)	Lowest; project-specific, audited	Credit issuance, finance, sale

Strategic implication for your funnel. The biggest jump in confidence is **Tier 1 → Tier 3**, and it is driven by **field measurement + methodology + verification**, not by satellite resolution. The most valuable paid step to offer a client is therefore a **professional pre-feasibility study with field validation**; higher-resolution imagery is a useful intermediate rung, not the destination.

9. What 180Climate can and cannot claim

Tool version	Can credibly claim	Must NOT claim
Free (Tier 1)	"Indicative, screening-grade avoided-emissions range ; suitable for go/no-go; order-of-magnitude."	A single number; a '% accuracy'; a creditable/bankable volume; a financial projection.
Higher-res (toward Tier 2)	"Improved pre-feasibility; tighter measurement; narrower range."	'Registry-grade'; high confidence on issued credits; that the baseline is resolved.

Tool version	Can credibly claim	Must NOT claim
Field-validated (Tier 3)	Registry-grade quantification — only after accredited methodology + independent verification.	Anything beyond what the validated methodology and verifier confirm.

In every case the user interface shows a **range, an uncertainty band, and the screening-grade qualifier**. This restraint is not a weakness — for a technical audience it is precisely what signals that the tool is built by people who understand carbon quantification.

10. Data-source architecture & vendor swapping

The tool is designed so the data source can be **swapped without rewriting the carbon engine**. The engine consumes a fixed internal data contract (forest-loss and biomass inputs with an uncertainty band) regardless of which vendor supplies it. Practically:

- Changing provider (free maps → commercial hi-res → LiDAR) is a **configuration / adapter change**, not a redesign.
- Each new vendor needs one small **adapter** to map its output into the tool's data contract — a bounded, well-defined task.
- Because commercial imagery is priced per scene/area, higher resolution is requested **only for flagged or shortlisted plots**, keeping cost controlled while preserving the option of a precision upgrade.

11. Assumptions

- Estimates are for **large concessions ($\geq 20,000$ ha)**, where aggregation reduces relative biomass error; small parcels carry proportionally higher uncertainty.
- The baseline reflects a **legally permitted harvest right being foregone** (HTI clear-fell or HA selective logging), consistent with the tool's permit-type routing; it is indicative and advisor-confirmed, not an ex-post validated baseline.
- Free Earth-observation products are assumed fit for **screening** only; quoted uncertainty ranges are typical literature values for tropical/Indonesian forest types and will vary by site, canopy, terrain, and cloud cover.
- Carbon-stock figures use standard biomass-to-CO₂e conversion and generic allometric/emission factors unless site data is supplied; this is appropriate for Tier 1 and is superseded by field measurement at Tier 3.
- All figures are **point-in-time** indicative estimates and are not adjusted for future market, policy, or permanence events.

12. Disclaimers

- **Indicative and screening-grade only.** Outputs are a first-pass estimate to support go/no-go decisions. They are **not** a registry-grade quantification and must not be used as the basis for issuing, forecasting, valuing, or trading carbon credits.
- **Not financial, investment, or legal advice.** Nothing here is a representation that any project will be eligible, registrable, or financially viable, or that any specific volume of credits will be issued.
- **Ranges, not guarantees.** Every carbon figure is an estimate with material uncertainty, expressed as a range and band. No single accuracy percentage is, or should be, claimed.
- **Confirmation required.** Any project decision should be confirmed by a full feasibility study including field measurement, an accredited methodology, and independent third-party validation/verification by an accredited body.
- **Data and methodology evolve.** Carbon standards, methodologies, eligibility rules, and the EUDR framework change; figures reflect the data and rules available at the date of this document.

13. Standards & references

1. IPCC (2006; 2019 Refinement) Guidelines for National GHG Inventories, Vol. 4 (AFOLU) — Tier framework and uncertainty.
2. GFOI — Integration of Remote-Sensing and Ground Data: Methods & Guidance Documentation (MGD).
3. CEOS / GFOI biomass and forest-cover-change good-practice guidance.
4. Verra VCS methodologies referenced by permit type: APD; IFM (VM0010 / VM0045); peat (VM0027, interim); and the AUD family (VM0048 / VMD0055) shown for contrast only.
5. West, T.A.P. et al. (2023) 'Action needed to make carbon offsets from forest conservation work', Science.
6. Guizar-Coutiño, A. et al. (2023) global evaluation of voluntary REDD+ effectiveness (Cambridge / VU Amsterdam).
7. Haya, B. et al. (2023) over-crediting assessment of REDD+ methodologies (UC Berkeley / Carbon Market Watch).
8. Synthesis of 44 REDD+ projects (2026) on over-crediting mechanisms, Nature Communications.
9. 52-project evaluation (2025) of tropical forest offsets, Science.
10. Santoro, M. & Cartus, O. — ESA CCI Biomass product and validation.
11. Dubayah, R. et al. — GEDI L4 aboveground biomass density and uncertainty.
12. Hansen, M.C. et al. — Global Forest Change; JRC Tropical Moist Forest; RADD (Wageningen) deforestation alerts.

14. Disciplines applied in this report

This assessment draws on, and is intended to be read against, the following forest-carbon and Earth-observation competencies — stated so a specialist reviewer can see the basis of the analysis:

- **Forest mensuration & allometry** — biomass and carbon-stock estimation from forest structure.
- **GHG inventory practice (IPCC Tiers)** — uncertainty classification and good-practice quantification.
- **REDD+ / VCS methodology** — baselines, additionality, leakage, permanence, and buffer accounting.
- **Optical & radar (SAR) remote sensing** — land-cover and forest-change detection (Sentinel-1/2, Landsat).
- **Spaceborne & airborne LiDAR** — canopy height and biomass calibration (GEDI, ICESat-2, ALS).
- **Uncertainty propagation & error analysis** — RMSE/%RMSE, confidence intervals, aggregation effects.
- **Additionality / baseline econometrics** — counterfactual and control-area methods underpinning credit integrity.

15. Take the next step with 180Climate

This screening estimate is the starting point, not the destination. Moving from an indicative Tier 1 figure to a financeable, registry-grade project takes expertise that 180Climate provides end to end:

- **Full pre-feasibility & feasibility study** — turning this screening range into a defensible, site-specific estimate.
- **Field measurement & validation** — ground plots and allometric sampling that move you from Tier 1 toward Tier 3, the single biggest gain in confidence.
- **Methodology & baseline advisory** — selecting and confirming the correct Verra methodology by permit type, and building a defensible, independently set baseline (the dominant driver of credit integrity).
- **Higher-resolution & LiDAR analysis** — narrowing the measurement uncertainty for shortlisted concessions.
- **EUDR compliance & export-readiness** — due-diligence statements, geolocation packs, and the 31 December 2020 deforestation-free assessment.
- **Project development, verification & market access** — guiding the project through accredited validation/verification and connecting it to finance and credit buyers.

Contact 180Climate to discuss your concession and the path from screening to a registry-grade, financeable project — info@180climate.net · www.180climate.net

Prepared as a methodology and limitations reference for the 180Climate Forest Carbon Pre-Feasibility Screening Tool. Figures are indicative and provided in good faith on the basis of openly available science and data; they carry the uncertainties described above and are subject to the disclaimers in Section 12.